Data Scientist | Data Analyst | Machine Learning Engineer

PORTOFOLIO

Presented by **Nikita Simanjuntak, S.Kom**



About Me

I'm a fresh graduate in Information Systems from Institut Teknologi Del, with a GPA of 3.27/4.00 and a strong interest in Data Analytics, Data Science, and Business Intelligence.

Throughout my academic journey, I developed hands-on experience in data processing, exploratory data analysis, machine learning, SQL, ETL, and dashboard development. I enjoy working with data to identify patterns, generate meaningful insights, and develop practical solutions to support better decision-making.

I'm currently looking for opportunities where I can apply my analytical and technical skills, contribute to real-world projects, and continue growing as a data professional.



@nikita_simanjuntak

nikitasimanjuntak

081397541049

nikitasimanjuntak005

Areas of Expertise

Business Intelligence

- Building dashboards and analytical solutions to support data-driven decision-making.

Data Analytics & Solutions

- Turning raw data into meaningful insights through SQL, data cleaning, exploratory data analysis, and data visualization.

Data Science

- Applying data processing, exploratory analysis, and machine learning approaches to solve data-driven problems.

Machine Learning

- Developing predictive models and data-driven solutions using Python and machine learning techniques.

Education

Bachelor of Information System
INSTITUT TEKNOLOGI DEL
GPA 3.27/4.00

2022 - 2026

Jalan Sisingamangaraja, Desa Sitoluama, Kecamatan Laguboti, Kabupaten Toba, Sumatera Utara, Kode Pos 22381





SKILLS

TECHNICAL SKILLS

- **Data & Analytics**: Data Cleaning · EDA · Statistical Analysis · Data Mining · Data Visualization · Predictive Modeling · Data Storytelling
- **Machine Learning & AI**: Machine Learning · Classification · Ensemble Learning · Recommendation Systems · Feature Engineering · Model Evaluation · Explainable AI
- **Database & Data Engineering**: Database Design · SQL Querying · Data Warehousing · ETL · Data Integration
- **Systems & Development**: Systems Analysis · Requirements Analysis · Web Development · REST API · Software Engineering · Software Testing

Software & Platforms

- Microsoft SQL Server
- MongoDB
- Tableau
- Google Colab
- Jupyter Notebook
- Figma
- Katalon
- Odoo
- Dolibarr
- VS Code
- Linux

Language

- Indonesian — Native
- English — Intermediate

Tools

- Github
- Git
- Drawio
- RapidMiner
- Pentaho
- Trello
- Flowgorithm

Programing Language

- C
- JavaScript
- Phyton
- SQL

EXPERIENCE

Internship

Software Engineer Intern
PT Telkom Indonesia – ROC-1 / IOC-R1 Medan
Juni - August 2025



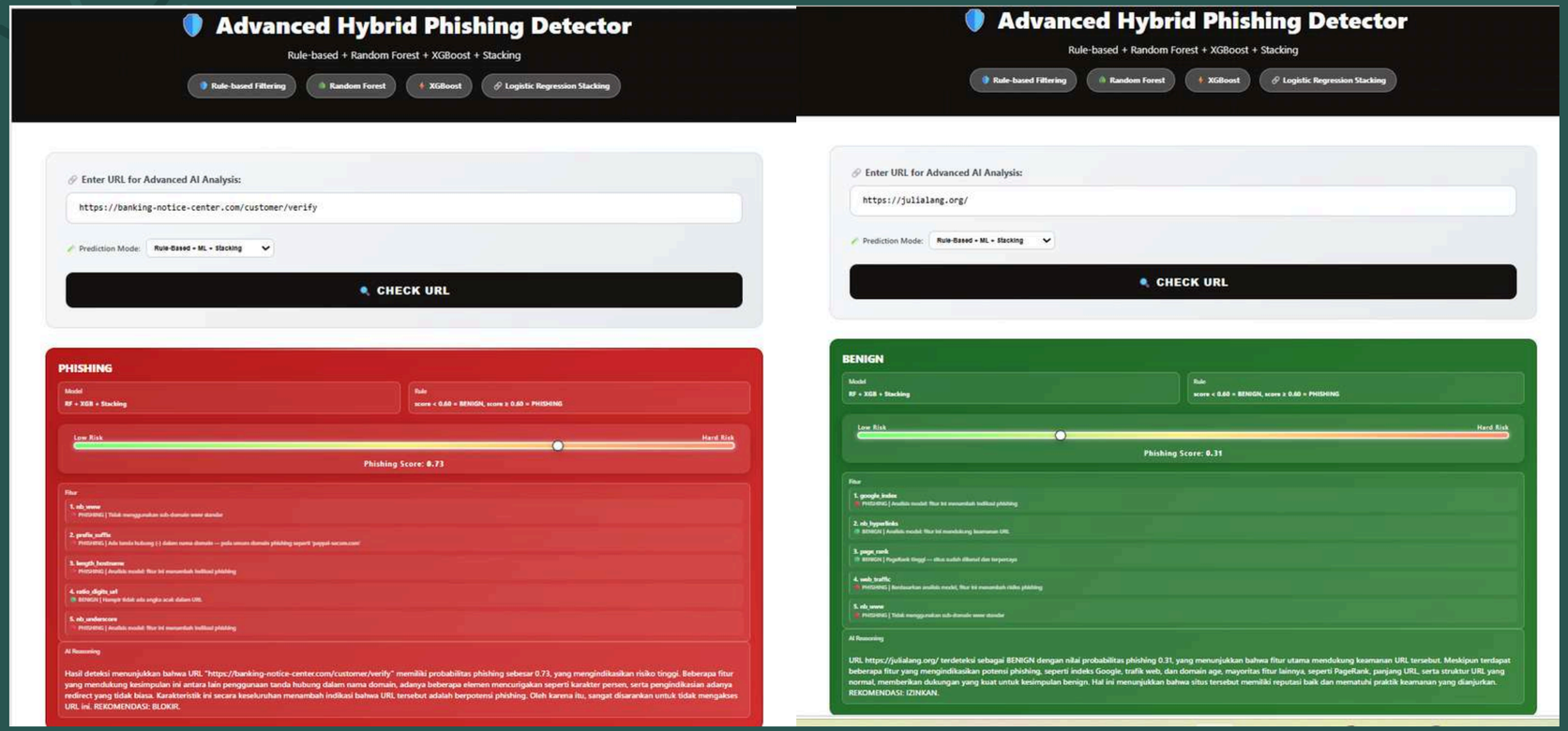
Recurring Wi-Fi Disruption Monitoring System
Developed a web-based monitoring information system for Telkom Infrastructure Indonesia (TIF) in the Sumatra region to monitor and track recurring Wi-Fi disruptions. The system provides centralized monitoring of Access Point status, disruption records, data integration, and reporting, helping operational teams identify issues and improve response efficiency.



Github

monitoring-jaringan-roc1

Final Project



Link Github

Methods & Algorithms

- Random Forest
- XGBoost
- Logistic Regression
- Stacking Ensemble
- Rule-Based Filtering

Key Features

- Rule-Based Filtering
- Stacking Ensemble
- SHAP Explainability
- LLM-based Reasoning

"**Hybrid Stacking Ensemble Integrated with Rule-Based Filtering for Phishing Website Detection: A Performance and Explainability Study.**

Developed a Hybrid Machine Learning-based phishing website detection system by integrating Rule-Based Filtering with a Stacking Ensemble (Random Forest + XGBoost + Logistic Regression). The system incorporates SHAP Explainability and LLM-based Reasoning to provide predictions along with insights into the factors influencing each model decision.

Projects

Return on Investment in Global Healthcare

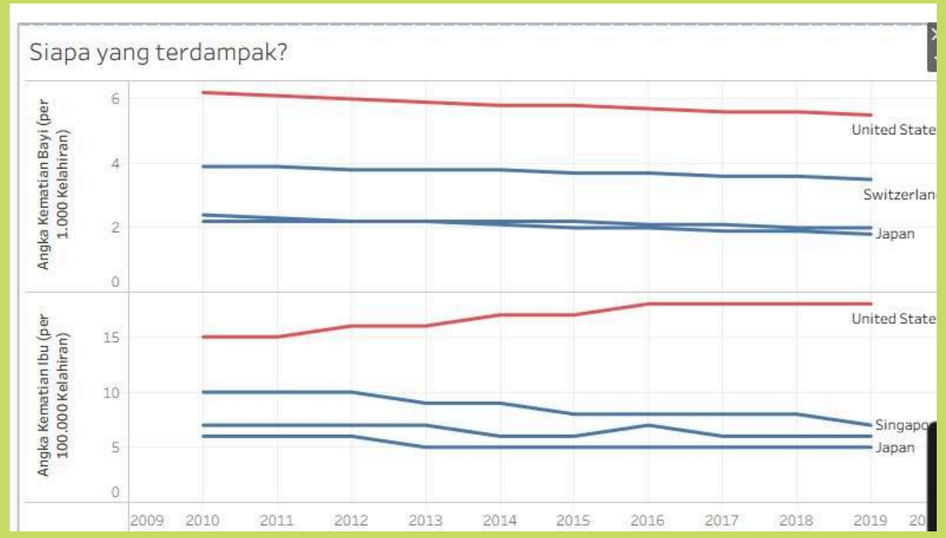
Developed an interactive Tableau data story analyzing healthcare spending, life expectancy, and investment efficiency across countries to identify spending inefficiencies and uncover best-performing healthcare systems.

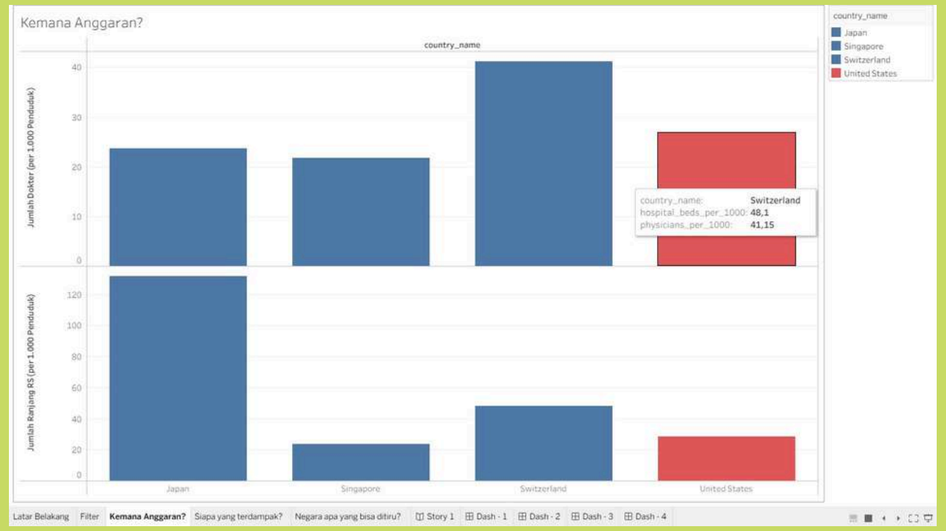
Focus
Healthcare Spending
Life Expectancy
ROI
Efficiency

Tool
Tableau

Github

Data Visualization



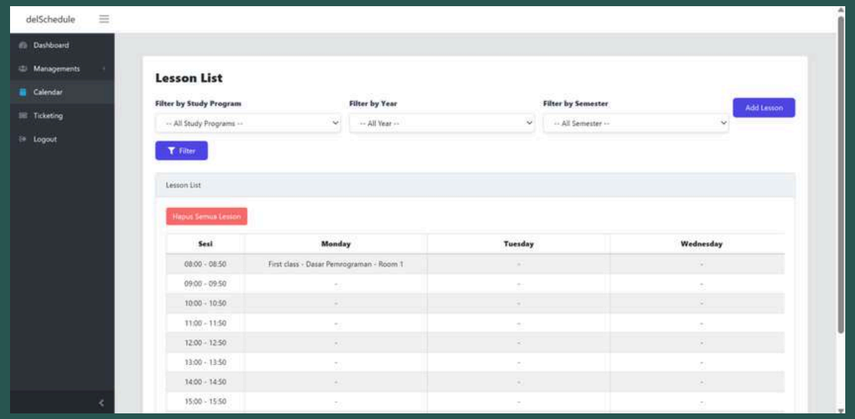


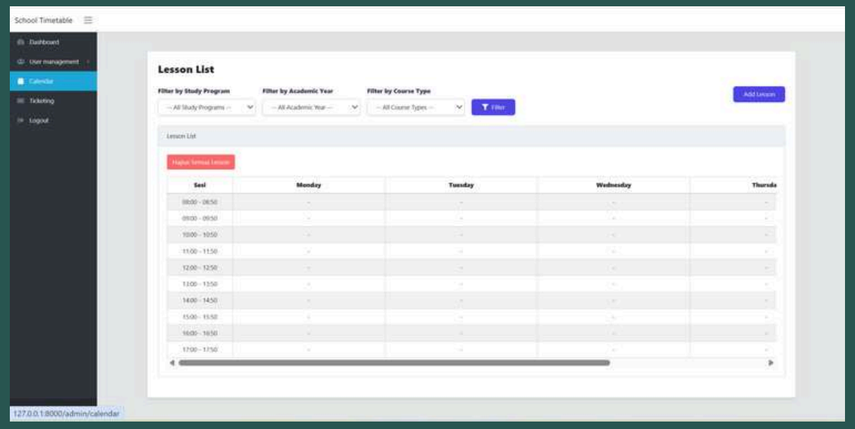
Projects

Information System Project

Course Scheduling Information System — DelSchedule

Developed a web-based course scheduling system to replace manual spreadsheet-based scheduling at Institut Teknologi Del. The system automates schedule management through an interactive calendar, filtering and search, CRUD operations, and ticketing for schedule change requests. Built using Laravel with MVC architecture and Role-Based Access Control (RBAC) to provide efficient and accessible academic scheduling.





Key Features

- 📅 Interactive Schedule Calendar — View schedules by semester, academic year, cohort, and study program.
- 🔎 Search & Filter — Quickly find courses, lecturers, rooms, and schedules.
- ⚙️ Schedule Management — Create, view, update, and delete academic schedules.
- 🎫 Ticketing System — Submit and manage schedule-change requests between students, lecturers, and academic administration.
- 🔐 Role-Based Access Control — Different access permissions based on user roles.

Github

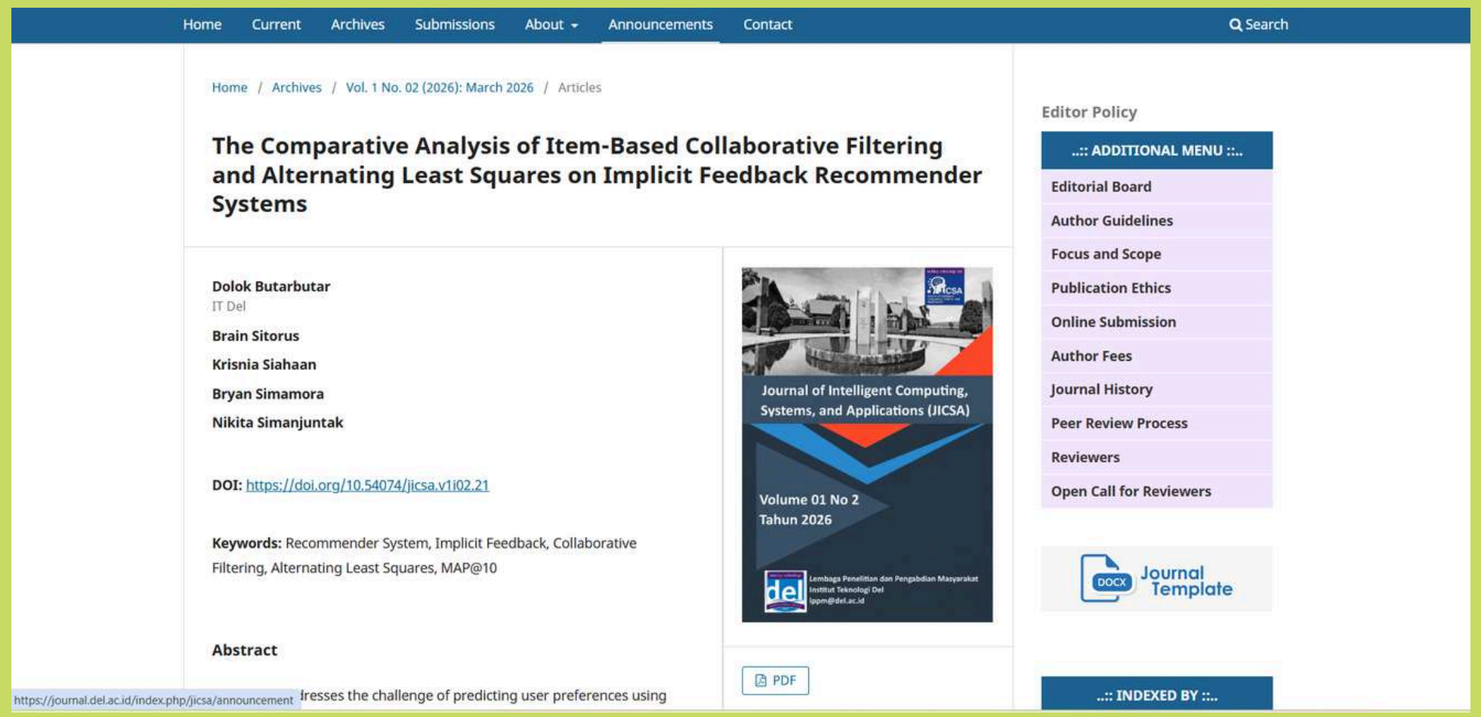
Projects

Recommendation System

Comparative Analysis of Item-Based Collaborative Filtering and Alternating Least Squares on Implicit Feedback Recommender Systems

Github

Publish Journal



Project Overview
Developed and evaluated an implicit-feedback recommender system to compare the performance of Item-Based Collaborative Filtering (IBCF) and Alternating Least Squares (ALS) in predicting user preferences.

Objective
To determine which collaborative filtering approach provides better Top-10 recommendation performance and assess the impact of hyperparameter tuning on ALS performance.

Projects

Data Warehouse & Business Intelligence

Data Mart-Based Dashboard for Student Admission Analysis

- **Project Overview**

Developed a Data Warehouse and Business Intelligence solution to integrate fragmented student admission data from multiple selection pathways (PMDK, USM1, USM2, USM3, and UTBK) into a centralized Data Mart and interactive dashboard. The system transforms raw Excel data into structured analytical information to support data-driven admission analysis and decision-making.

- **Objective**

To provide an integrated analytical platform for identifying applicant trends, study program preferences, geographic distribution, and admission outcomes.

Key Features

- Admission Overview — Analyze total applicants and admission outcomes.
- Study Program Analysis — Identify program preferences and applicant distribution.
- Geographic Analysis — Visualize applicant distribution by region.
- Selection Path Analysis — Compare PMDK, USM, and UTBK admission pathways.
- Trend Analysis — Monitor applicant and admission trends across periods.

Multiple Excel Sources

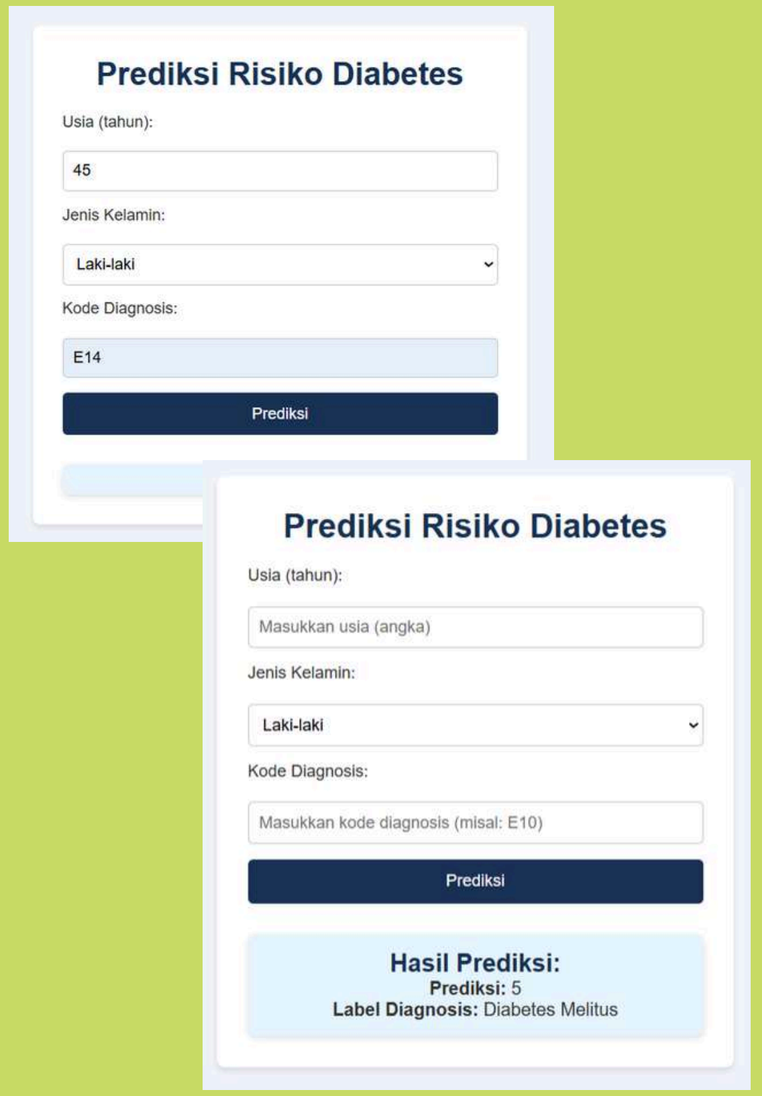
→ ETL (Extract, Transform, Load)
→ Dimensional Data Mart
→ Interactive BI Dashboard
→ Admission Insights

Github

Projects

Data Mining

Application of Support Vector Machine Model to Predict Diabetes Mellitus Diagnosis and Verification Cost Estimation for BPJS Users



Project Overview
Developed a machine learning classification model to identify individuals at higher risk of Diabetes Mellitus based on demographic, healthcare access, and health-related factors. The project applies Support Vector Machine (SVM) to uncover patterns in patient data and support data-driven risk identification.

Objective
To develop a predictive model capable of classifying individuals based on their risk of Diabetes Mellitus using relevant demographic and healthcare-related features.

Github

CERTIVICATE



14773720

BADAN NASIONAL
SERTIFIKASI PROFESI
INDONESIAN PROFESSIONAL CERTIFICATION AUTHORITY

SERTIFIKAT KOMPETENSI
CERTIFICATE OF COMPETENCE

No. 63111 2511 5 0162903 2025

Dengan ini menyatakan bahwa,
This is to certify that,

Nikita Simanjuntak

No. Reg. DMI 1565 47592 2025

Telah kompeten pada bidang:
Is competent in the area of:

Sains Data
Data Science

Dengan Kualifikasi / Kompetensi:
With Qualification / Competency:

Ilmuwan Data Muda
Associate Data Scientist

Sertifikat ini berlaku untuk: 3 (tiga) tahun
This certificate is valid for: 3 (three) years

Yogyakarta, 17 Desember 2025
Atas Nama Badan Nasional Sertifikasi Profesi
On Behalf of Indonesian Professional Certification Authority

Lembaga Sertifikasi Profesi Teknologi Digital
Professional Certification Body of Digital Technology

LSP TEKNOLOGI DIGITAL

Ir. Gunawan Ramli, M.Kom.
(Direktur / Director)

BNSP

Daftar Unit Kompetensi
List of Competency Units

No.	Kode Unit Kompetensi *Code of Competency Unit*	Judul Unit Kompetensi *Title Of Competency Unit*
1.	J.62DMI00.004.1	Mengumpulkan Data *Collecting Data*
2.	J.62DMI00.005.1	Menelaah Data *Examining Data*
3.	J.62DMI00.006.1	Memvalidasi Data *Validating Data*
4.	J.62DMI00.007.1	Menentukan Objek Data *Defining Data Objects*
5.	J.62DMI00.008.1	Membersihkan Data *Cleaning Data*
6.	J.62DMI00.009.1	Mengkonstruksi Data *Constructing Data*
7.	J.62DMI00.010.1	Menentukan Label Data *Determining Data Labels*
8.	J.62DMI00.013.1	Membangun Model *Building Models*
9.	J.62DMI00.014.1	Mengevaluasi Hasil Pemodelan *Evaluating Modeling Results*

Yogyakarta, 17 Desember 2025
Atas Nama Badan Nasional Sertifikasi Profesi
On Behalf of Indonesian Professional Certification Authority
Lembaga Sertifikasi Profesi Teknologi Digital
Professional Certification Body of Digital Technology

3 X 4

PT TEKNOLOGI DIGITAL

Nikita Simanjuntak
Tanda tangan Pemilik
(Signature of Holder)

Roy Buana
Manager Sertifikasi
(Manager of Certification)

BNSP

Associate Data Scientist
BNSP – LSP Teknologi Digital

Vocational School Graduate Academy (VSGA) · Digital Talent Scholarship · 2025

A Certified in data integration, preprocessing, analytical modeling, and model evaluation through a practical task-based assessment using RapidMiner.

ORGANIZATION



SERTIFIKAT KEANGGOTAAN

KELUARGA BESAR MAHASISWA
INSTITUT TEKNOLOGI DEL
Menyatakan bahwa

Nikita Simanjuntak

Telah menjadi
Anggota
Departemen Agama dan Sosial
Badan Eksekutif Mahasiswa Keluarga Besar Mahasiswa Institut Teknologi Del Pada Periode Kepengurusan TA 2023/2024 .

Yoke April Lia Purba
Koordinator Kemahasiswaan

Rony Fritz N. Bakara
Ketua BEM IT Del 2023/2024

BEM IT Del (2023–2024) – Member, Department of Religion and Social Affairs Contributed to the execution of community service programs, student engagement events, and spiritual initiatives within the campus community.



SERTIFIKAT KEANGGOTAAN
KELUARGA BESAR MAHASISWA INSTITUT TEKNOLOGI DEL

MENYATAKAN BAHWA

Nikita Simanjuntak
12S22005 - S1 Sistem Informasi

Telah menjadi **Bendahara**
Departemen Agama dan Sosial
Badan Eksekutif Mahasiswa Keluarga Besar Mahasiswa Institut Teknologi Del Kabinet Sahat Marsada Periode 2024/2025

Yoke April Lia Purba, S.Kom
KEPALA BIDANG KEMAHASISWAAN

Samuel Janring Sitio
KETUA BEM IT DEL PERIODE 2024/2025

Hebert Timony Aritonang
KEPALA DEPARTEMEN AGAMA DAN SOSIAL

BEM IT Del (2024–2025) – Treasurer, Department of Religion and Social Affairs Managed financial planning, budgeting, and financial reporting for departmental social initiatives and religious activities under Kabinet Sahat Marsada.

VOLUNTEER



SERTIFIKAT

PENGHARGAAN

Dengan bangga diberikan kepada:

NIKITA SIMANJUNTAK

Atas kontribusinya sebagai :

ANGGOTA SEKSI ACARA

Panitia Kaderisasi Sistem Informasi Angkatan 2023

Sitoluama, 09 Maret 2024

Samuel Indra Gunawan Situmeang, S.TI.,M.Sc

Ketua Program Studi Sarjana Sistem Informasi Institut Teknologi Del



Institut Teknologi Del

Institut Teknologi Del, Desa Sitoluama, Kec. Laguboti, Kab. Toba, Sumatera Utara, 22381, Indonesia
ph. +62 (632) 331234. Fax. +62 (632) 331116, www.del.ac.id, email : info@del.ac.id

CERTIFICATE

OF APPRECIATION

THE FOLLOWING AWARD IS GIVEN TO

Nikita Simanjuntak

For participation as a **member of the Decoration Division** in The Inauguration, Dies Natalis & Graduation Ceremony at Institut Teknologi Del
Sitoluama, Sumatera Utara, Indonesia, 22 - 23 September 2023

Dr. Arnaldo Marulitua Sinaga, S.T., M.InfoTech
Rector of Institut Teknologi Del

Dr. Johannes Harunggungan Sianpar, S.T., M.T.
Head of Steering Commitee

Josua Boyke William Jawak , S.T., M.Ds
Head of Organizing Commitee

CERTIVICATE



IOE A powerful and balanced voice for business

CERTIFICATE
OF MODULE COMPLETION

THIS CERTIFIES THAT

Nikita Simanjuntak

has successfully completed **Course 1** of the **IOE-Microsoft AI Fluency Training**. This course provided foundational knowledge of artificial intelligence, introducing key concepts, terminology, and the underlying principles that define the field. By completing this course, the participant has taken the first step toward building a comprehensive understanding of AI and its potential to transform the world of work and society at large.

APINDO

March 11, 2026



freeCodeCamp

This certifies that

nikitasimanjuntak

successfully completed the

Machine Learning with Python

Developer Certification on March 14, 2026

representing approximately 300 hours of work

Quincy Larson
Executive Director, freeCodeCamp.org

Verify this certification at:
https://freecodecamp.org/certification/nikitasimanjuntak/machine-learning-with-python-v7



IOE A powerful and balanced voice for business Microsoft

CERTIFICATE
OF QUIZ COMPLETION

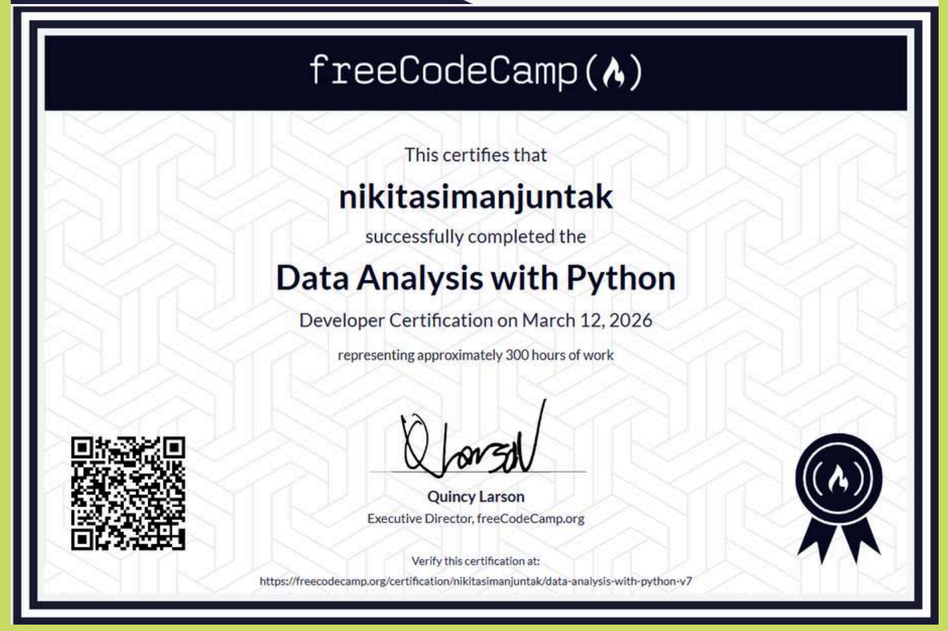
THIS CERTIFIES THAT

Nikita Simanjuntak

has successfully completed the **AI Fluency Training**, developed by the **International Organisation of Employers (IOE)** in collaboration with **Microsoft**. This comprehensive programme consists of six structured courses. Through this training, the participant has gained a solid foundation in artificial intelligence (AI), including: core AI concepts and terminology, real-world applications of AI across industries, responsible use of AI, the impact of AI on the future of work, strategies for business adoption of AI, and the development of an AI-ready mindset. By completing this programme, the participant is now equipped with the knowledge and awareness needed to actively engage with AI-driven transformation and contribute to shaping a responsible and inclusive AI landscape.

APINDO

March 11, 2026



freeCodeCamp

This certifies that

nikitasimanjuntak

successfully completed the

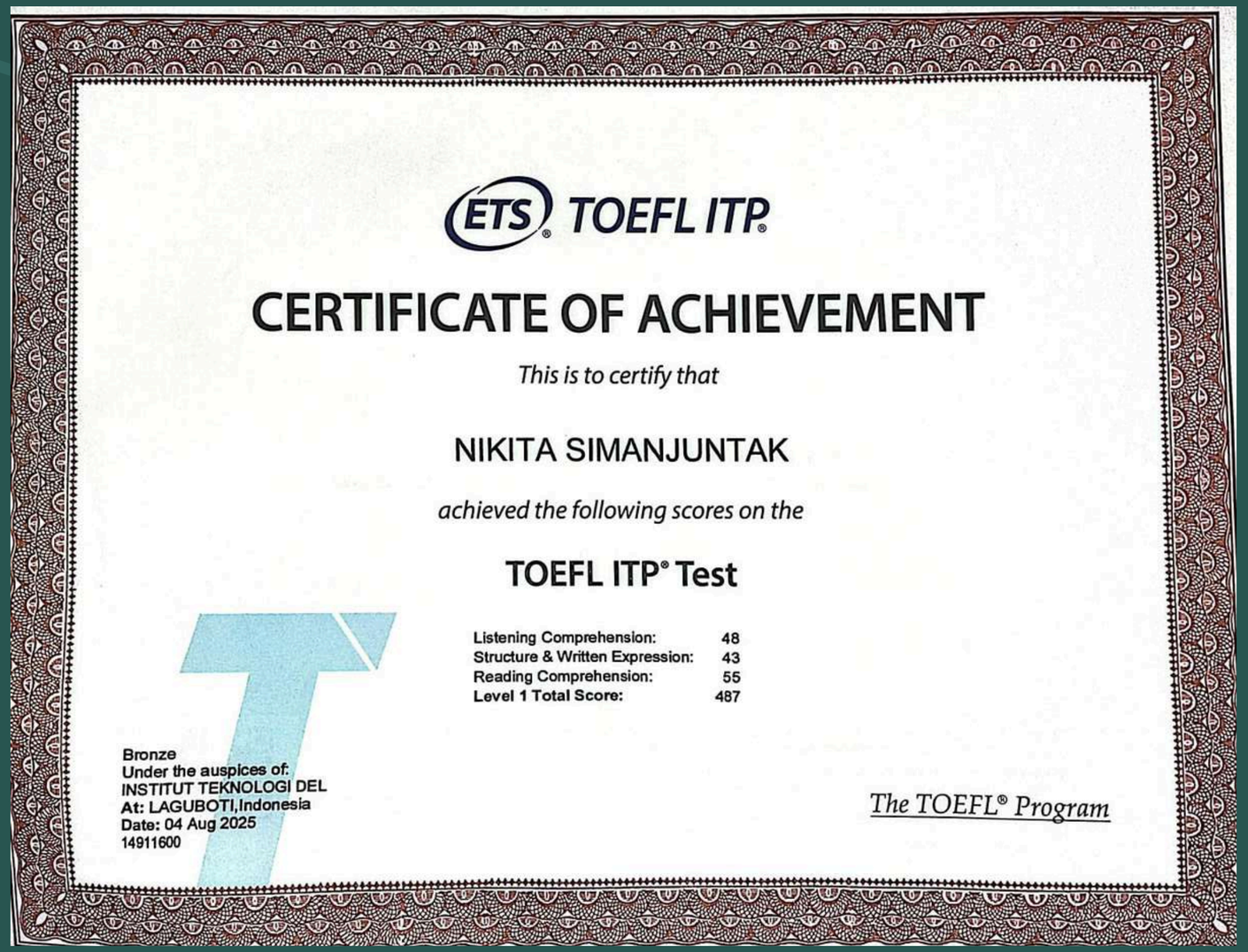
Data Analysis with Python

Developer Certification on March 12, 2026

representing approximately 300 hours of work

Quincy Larson
Executive Director, freeCodeCamp.org

Verify this certification at:
https://freecodecamp.org/certification/nikitasimanjuntak/data-analysis-with-python-v7



ETS TOEFL ITP

CERTIFICATE OF ACHIEVEMENT

This is to certify that

NIKITA SIMANJUNTAK

achieved the following scores on the

TOEFL ITP® Test

Listening Comprehension:	48
Structure & Written Expression:	43
Reading Comprehension:	55
Level 1 Total Score:	487

Bronze
Under the auspices of:
INSTITUT TEKNOLOGI DEL
At: LAGUBOTI,Indonesia
Date: 04 Aug 2025
14911600

The TOEFL® Program

Thank You